Agile Planning and Portfolio Management with Azure Boards

In this lab, you'll learn about the agile planning and portfolio management tools and processes provided by Azure Boards and how they can help you quickly plan, manage, and track work across your entire team. You'll explore the product backlog, sprint backlog, and task boards that can track the flow of work during an iteration. We'll also look at the enhanced tools in this release to scale for larger teams and organizations.

You will learn how to:

- Manage teams, areas, and iterations.
- Manage work items.
- Manage sprints and capacity.
- Customize Kanban boards.
- Define dashboards.
- Customize team process.

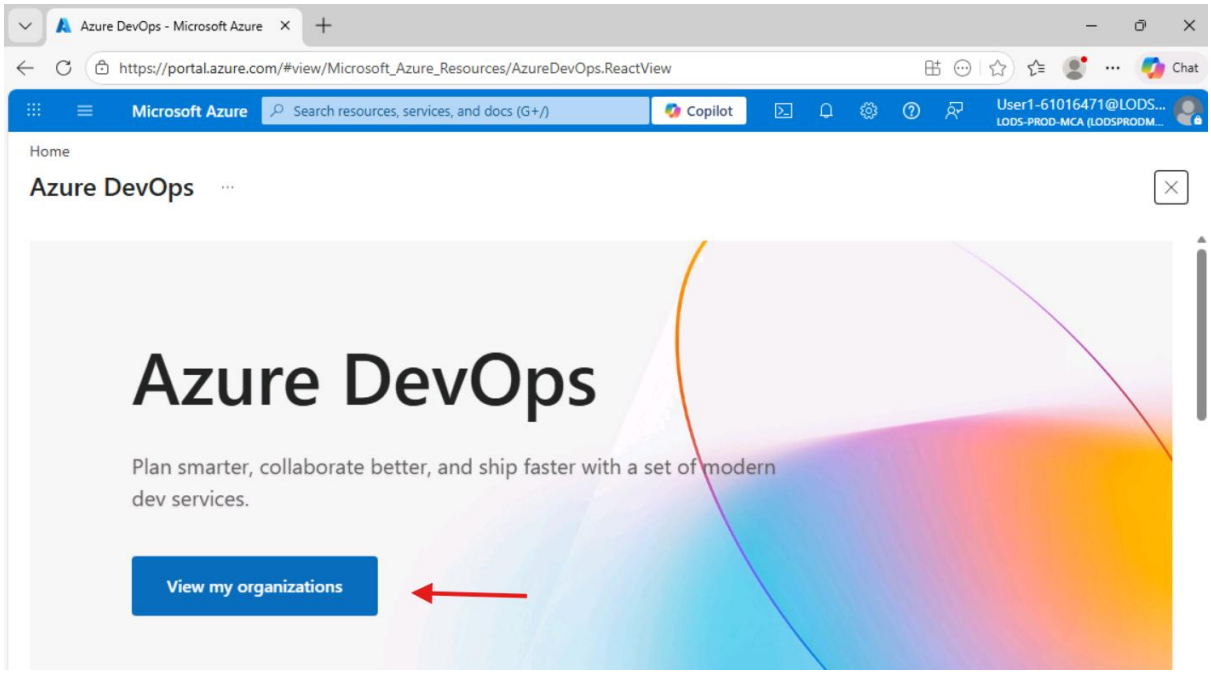
Before we start

To Prerequisites, you need:

- **Microsoft Edge** or an Azure DevOps supported browser.
- An active Azure subscription. If you don't have one, you can create a free account at Azure Free Account.
- An Azure DevOps organization. If you don't already have one, create one by following the instructions at Create an organization or project collection.

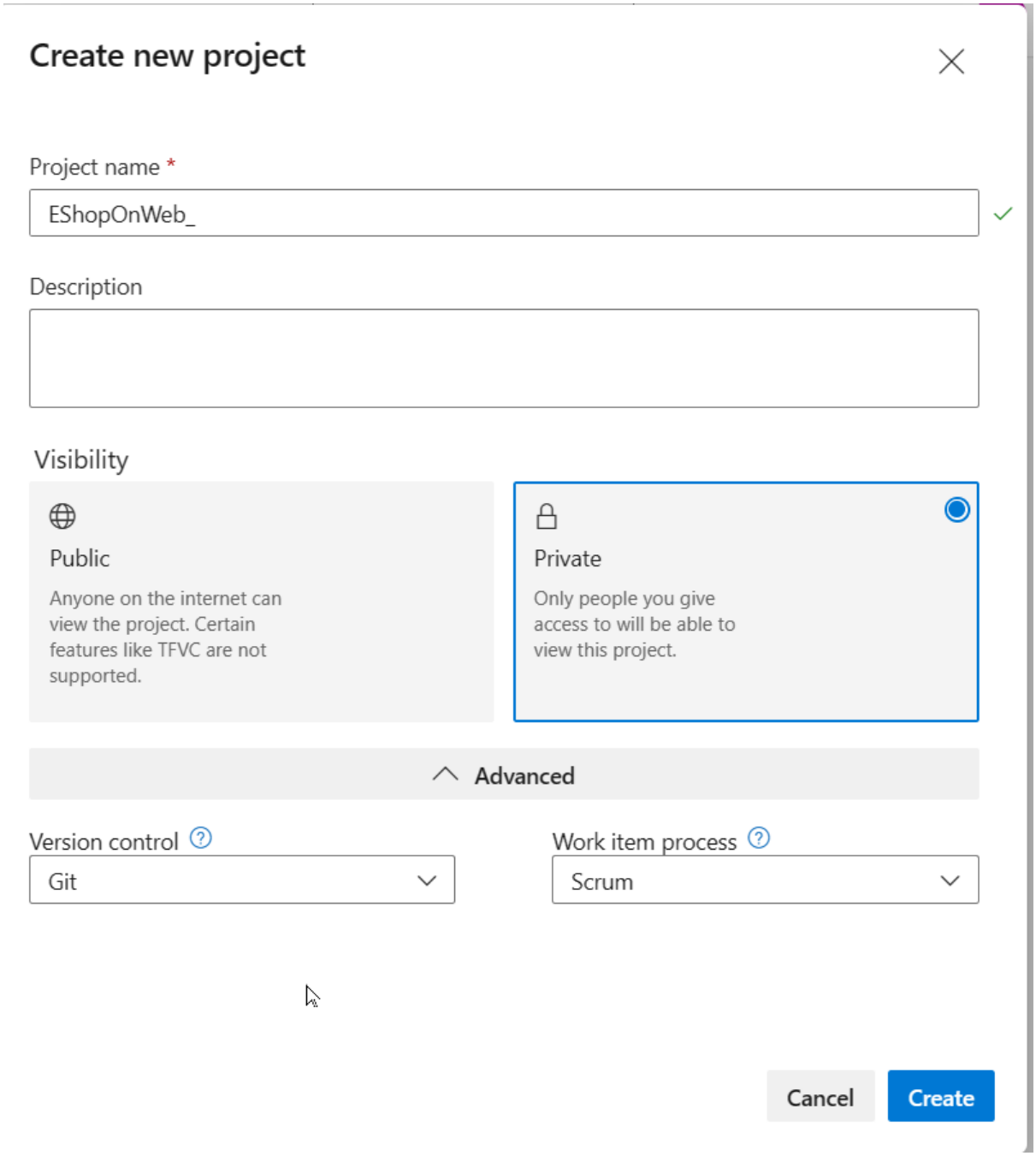
Open a browser and navigate to the Azure portal at https:*//portal.azure.com*, then search at the top of the Azure portal screen for **Azure DevOps**.

In the resulting page, select **Azure DevOps organizations**



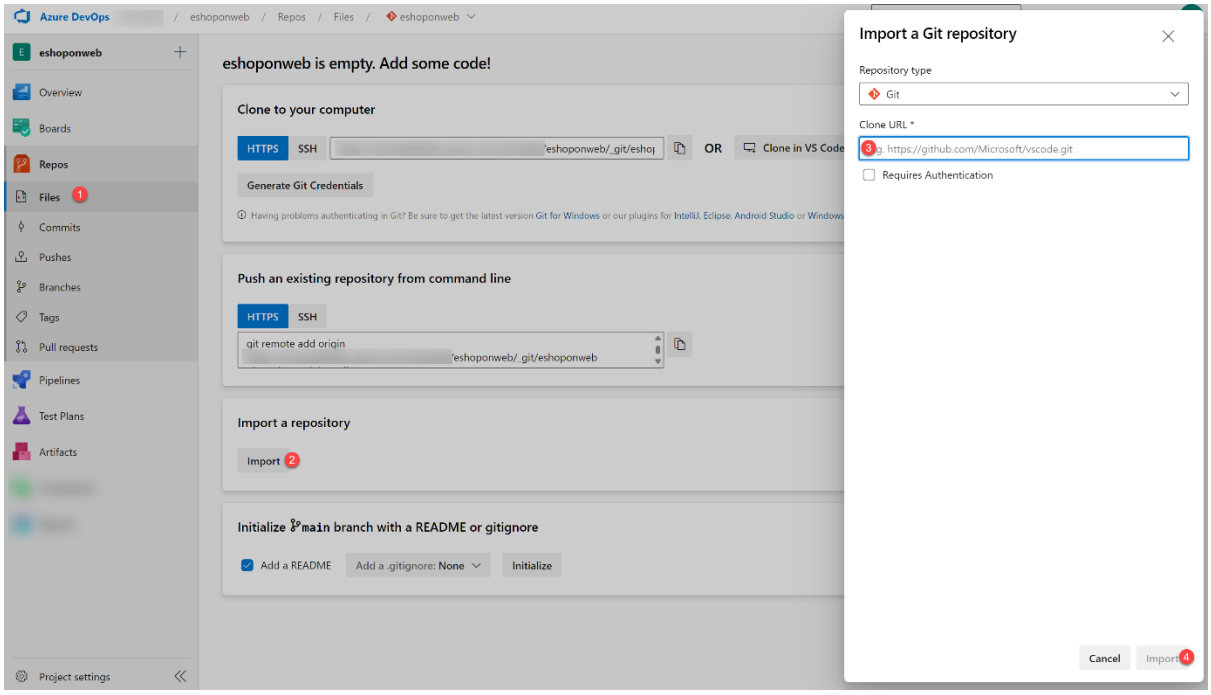
Next, select the link labelled **My Azure DevOps Organizations** or navigate directly to https://aex.dev.azure.com.

1. Open your browser and navigate to your Azure DevOps organization.
2. Select the **New Project** option and use the following settings:
    - name: **eShopOnWeb**
    - visibility: **Private**
    - Advanced: Version Control: **Git**
    - Advanced: Work Item Process: **Scrum**
3. Select **Create**.



Import the eShopOnWeb git repository

1. Open the previously created **eShopOnWeb** project.
2. Select the **Repos > Files**, **Import a Repository** and then select **Import**.
3. On the **Import a Git Repository** window, paste the following URL https:*//github.com/MicrosoftLearning/eShopOnWeb* and select **Import**:



4. The repository is organized the following way:
    - **.ado** folder contains Azure DevOps YAML pipelines.
    - **.devcontainer** folder contains setup to develop using containers (either locally in VS Code or GitHub Codespaces).
    - **.azure** folder contains Bicep & ARM infrastructure as code templates.
    - **.github** folder contains YAML GitHub workflow definitions.
    - **src** folder contains the .NET 8 website used on the lab scenarios.
5. Leave the web browser window open.
6. Go to **Repos > Branches**.
7. Hover on the **main** branch then select the ellipsis on the right of the column.
8. Select **Set as default branch**.

Manage teams, areas, and iterations

Each new project is configured with a default team, which name matches the project name. You have the option of creating additional teams. Each team can be granted access to a suite of Agile tools and team assets. The ability to create multiple teams gives you the flexibility to choose the proper balance between autonomy and collaboration across the enterprise.

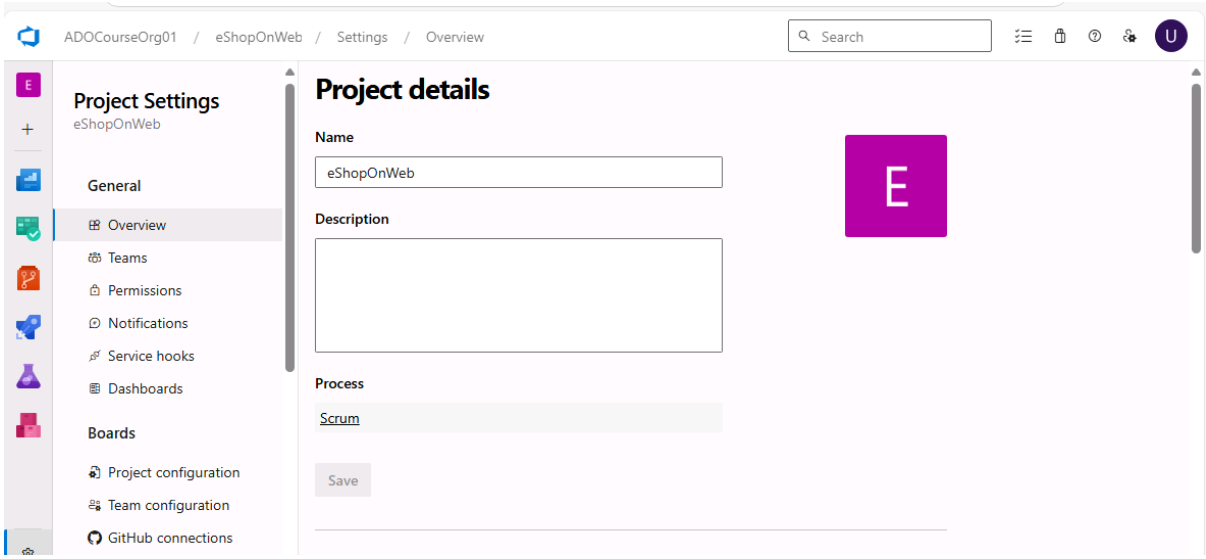
1. On your lab computer, start a web browser and navigate to the Azure DevOps portal at https:*//aex.dev.azure.com*.

Note: If prompted, sign in using the Microsoft account associated with your Azure DevOps subscription.

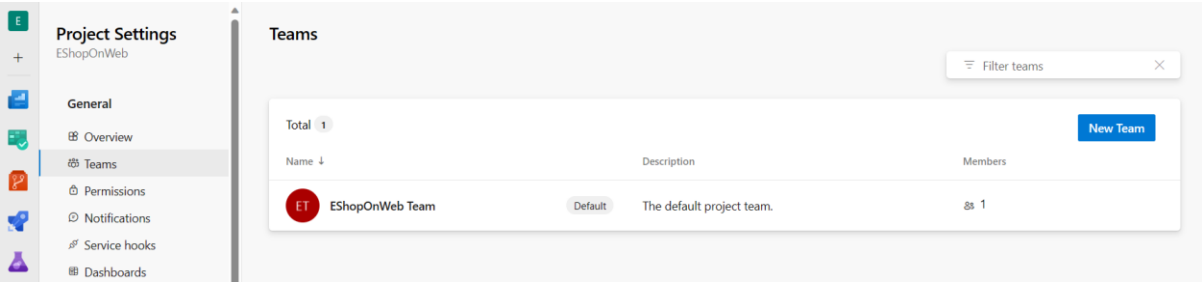
2. Open the **eShopOnWeb** project under your Azure DevOps organization.

Note: Alternatively, you can access the project page directly by navigating to the https://dev.azure.com/YOUR-AZURE-DEVOPS-ORGANIZATION/PROJECT-NAME URL, where the YOUR-AZURE-DEVOPS-ORGANIZATION placeholder, represents your account name, and the PROJECT-NAME placeholder represents the name of the project.

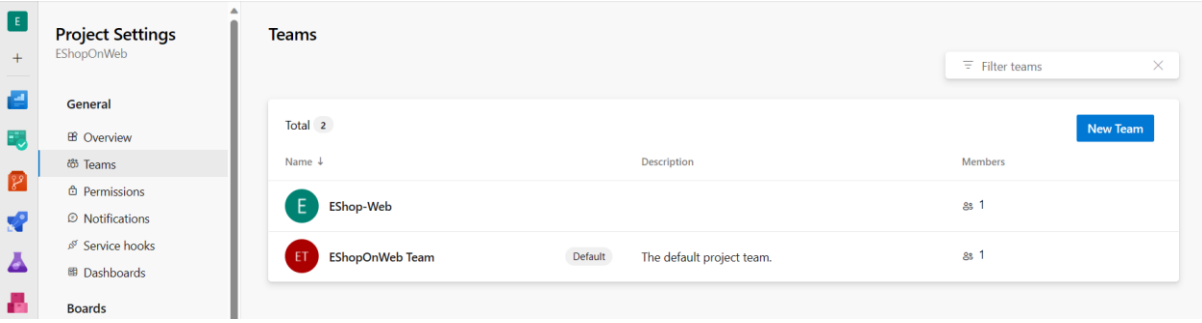
3. Select the cogwheel icon labeled **Project settings** located in the lower left corner of the page to open the **Project settings** page.



4. In the **General** section, select the **Teams** tab. There is already a default team in this project, **eShopOnWeb Team** but you'll create a new one for this lab. Select **New Team**.



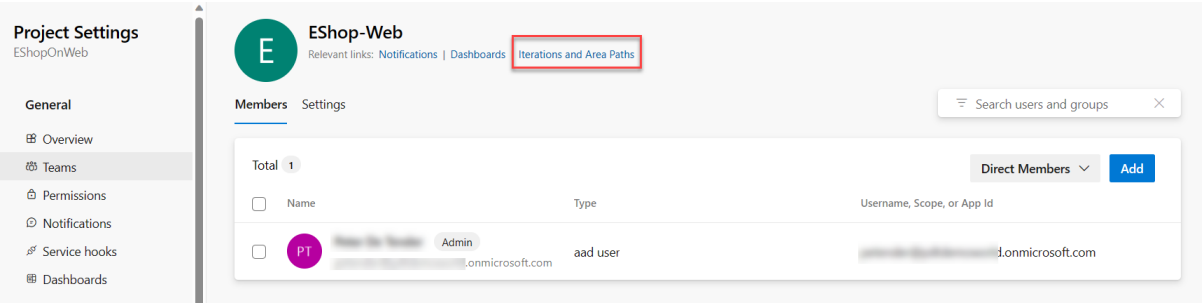
5. On the **Create a new team** pane, in the **Team name** textbox, type **EShop-Web**, leave other settings with their default values, and select **Create**.



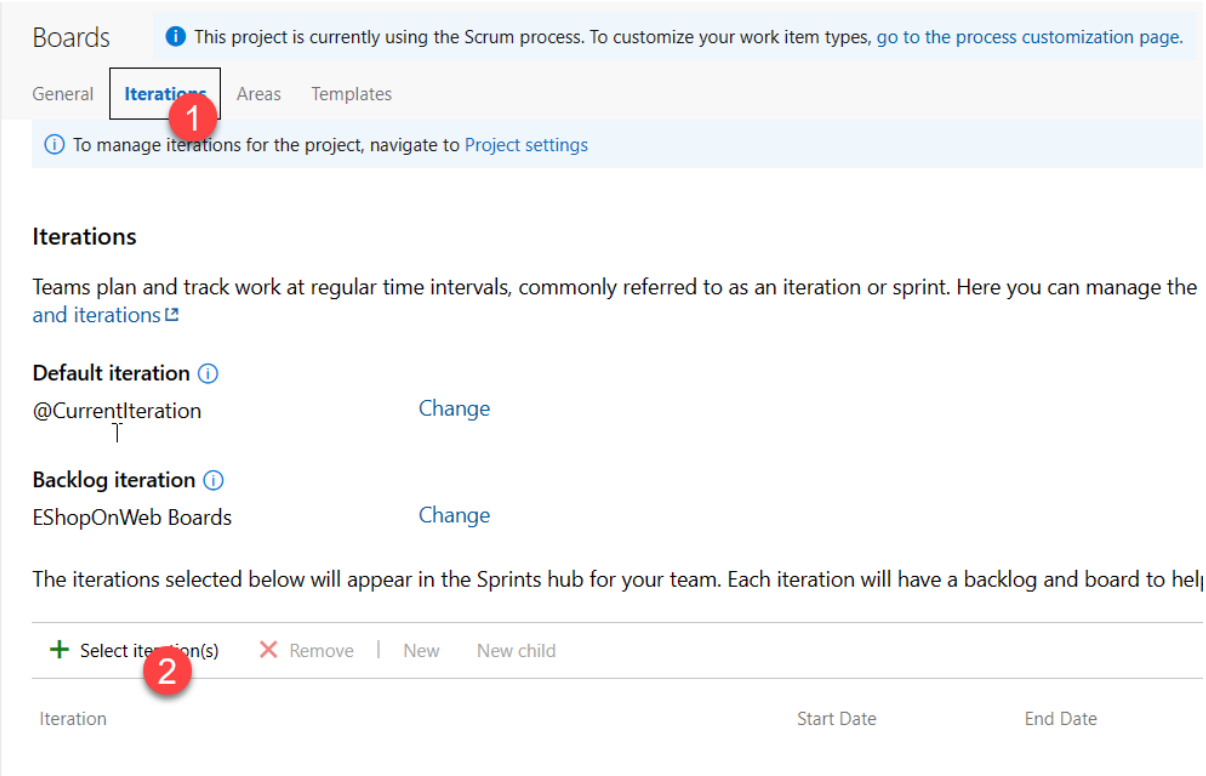
6. In the list of **Teams**, select the newly created team to view its details.

Note: By default, the new team has only you as its member. You can use this view to manage such functionality as team membership, notifications, and dashboards.

7. Select **Iterations and Area Paths** link at the top of the **EShop-Web** page to start defining the schedule and scope of the team.



8. At the top of the **Boards** pane, select the **Iterations** tab and then select **+ Select iteration(s)**.

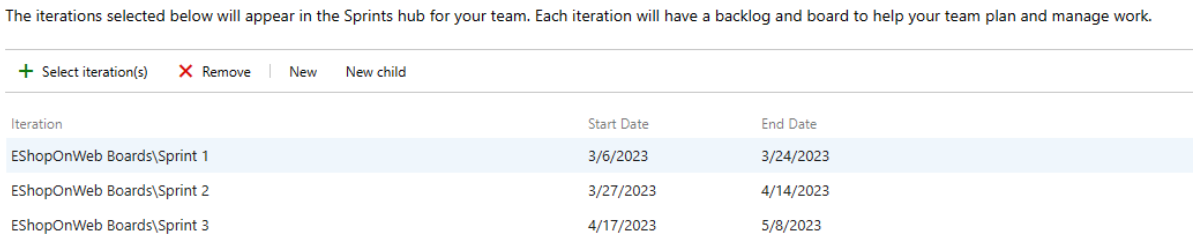


9. Select **eShopOnWeb\Sprint 1** and select **Save and close**. Note that this first sprint will show up in the list of iterations, but the Dates are not set yet.

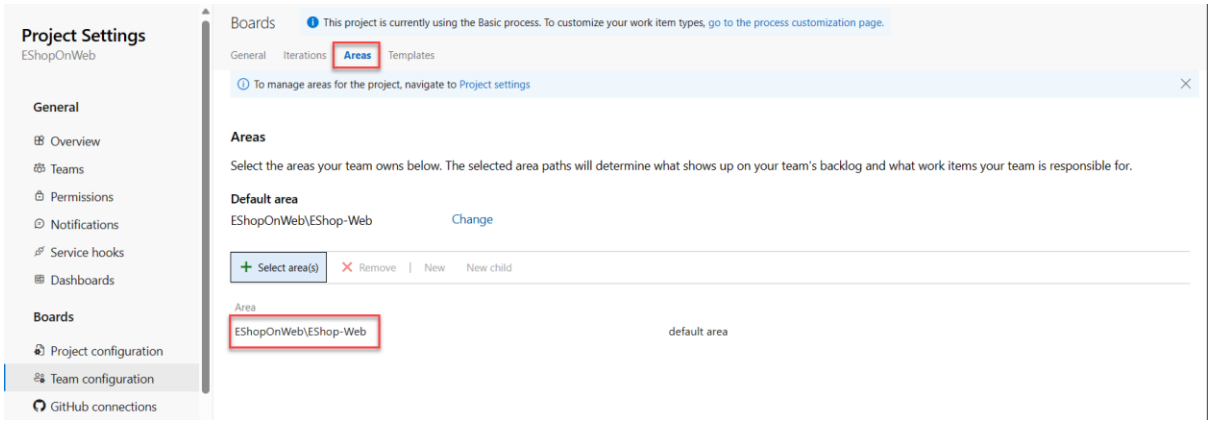
10. Select **Sprint 1** and select the **ellipsis (…)**. From the context menu, select **Edit**.

Note: Specify the Start Date as the first work day of last week, and count 3 full work weeks for each sprint. For example, if March 6 is the first work day of the sprint, it goes until March 24th. Sprint 2 starts on March 27, which is 3 weeks out from March 6.

11. Repeat the previous step to add **Sprint 2** and **Sprint 3**. You could say that we are currently in the 2nd week of the first sprint.



12. Still in the **Project Settings / Boards / Team Configuration** pane, at the top of the pane, select the **Areas** tab. You will find there an automatically generated area with the name matching the name of the team.



13. Select the ellipsis symbol (...) next to the **default area** entry and, in the dropdown list, select **Include sub areas**.

Note: The default setting for all teams is to exclude sub-area paths. We will change it to include sub-areas so that the team gets visibility into all of the work items from all teams. Optionally, the management team could also choose to not include sub-areas, which automatically removes work items from their view as soon as they are assigned to one of the teams.

Manage work items

Work items play a prominent role in Azure DevOps. Whether describing work to be done, impediments to release, test definitions, or other key items, work items are the workhorse of modern projects. In this task you'll focus on using various work items to set up the plan to extend the eShopOnWeb site with a product training section. While it can be daunting to build out such a substantial part of a company's offering, Azure DevOps and the Scrum process make it very manageable.

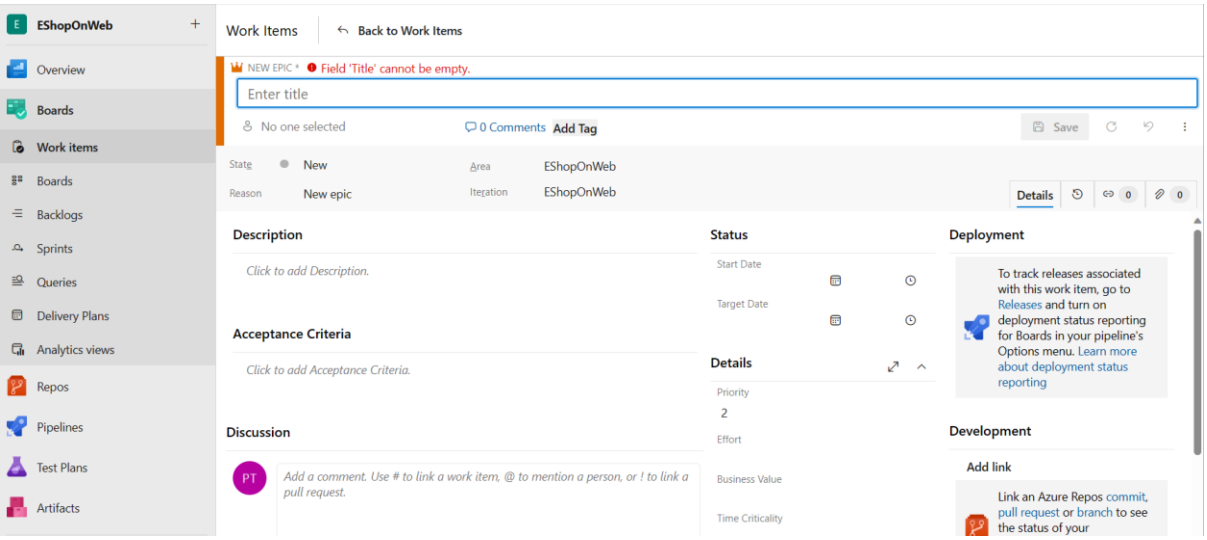
Note: This task is designed to illustrate a variety of ways you can create different kinds of work items, as well as to demonstrate the breadth of features available on the platform. As a result, these steps should not be viewed as prescriptive guidance for project management. The features are intended to be flexible enough to fit your process needs, so explore and experiment as you go.

1. Select the project name in the upper left corner of the Azure DevOps portal to return to the project home page.

2. In the vertical navigational pane of the Azure DevOps portal, select the **Boards** icon and, select **Work Items**.

Note: There are many ways to create work items in Azure DevOps, and we'll explore a few of them. Sometimes it's as simple as firing one off from a dashboard.

3. On the **Work Items** window, select **+ New Work Item > Epic**.



4. In the **Enter title** textbox, type **Product training**.

5. In the upper left corner, select the **No one selected** entry and, in the dropdown list, select your user account in order to assign the new work item to yourself. If your name doesn't appear to start with, begin typing your name and select **Search**.

6. Next to the **Area** entry, select the **eShopOnWeb** entry and, in the dropdown list, select **EShop-Web**. This will set the **Area** to **eShopOnWeb\EShop-Web**.

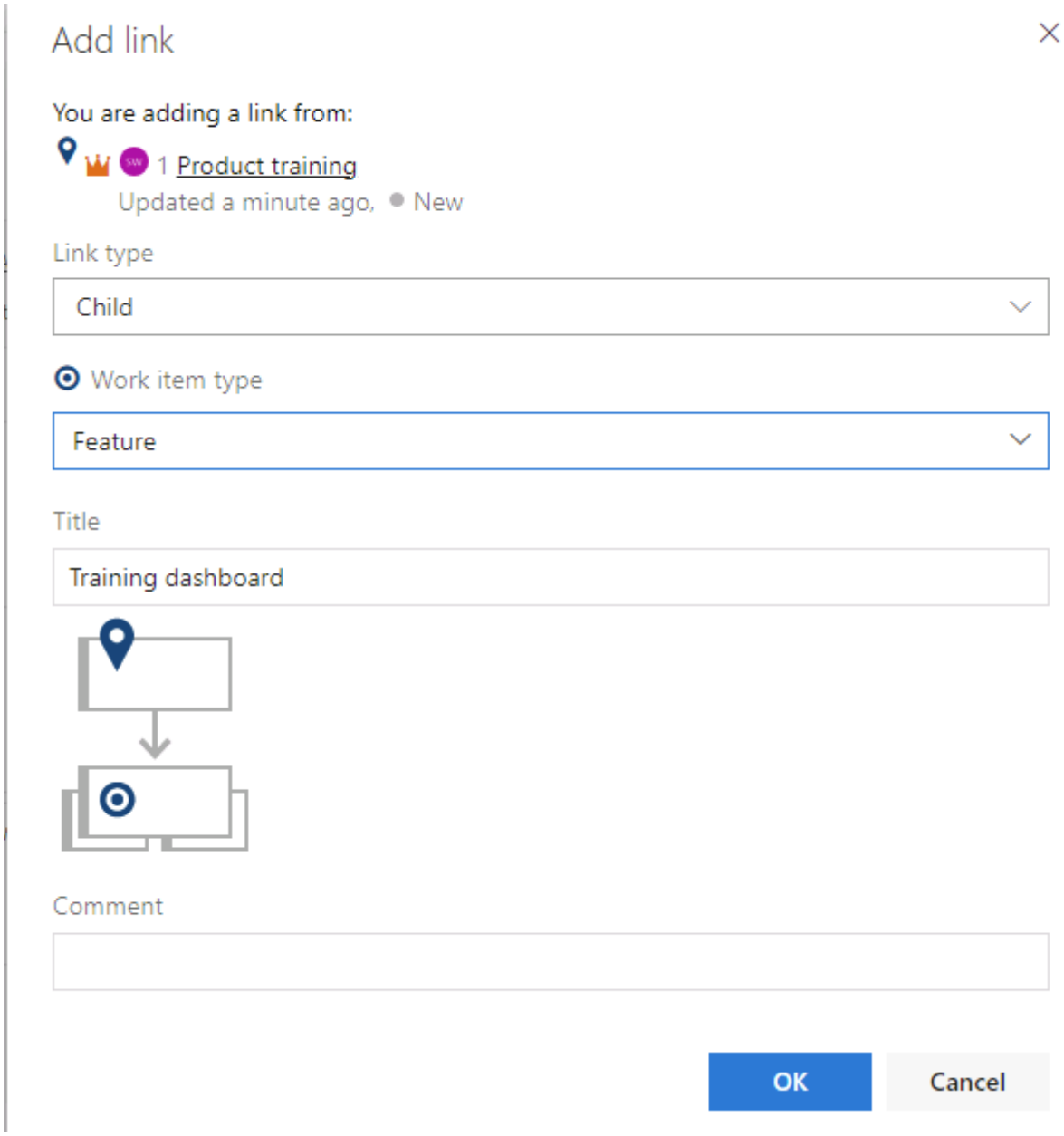
7. Next to the **Iteration** entry, select the **eShopOnWeb** entry and, in the dropdown list, select **Sprint 2**. This will set the **Iteration** to **eShopOnWeb\Sprint 2**.

8. Select **Save** to finalize your changes. **Do not close it**.

Note: The work item form includes all of the relevant work item settings. This includes details about who it's assigned to, its status across many parameters, and all the associated information and history for how it has been handled since creation. One of the key areas to focus on is the **Related Work**. We will explore one of the ways to add a feature to this epic.

9. In the **Related work** section on the lower right-side, select the **Add link** entry and, in the dropdown list, select **New item**.

10. On the **Add link** panel, in the **Link Type** dropdown list, select **Child**. Next, in the **Work item type** dropdown list, select **Feature**, in the **Title** textbox, type **Training dashboard**.



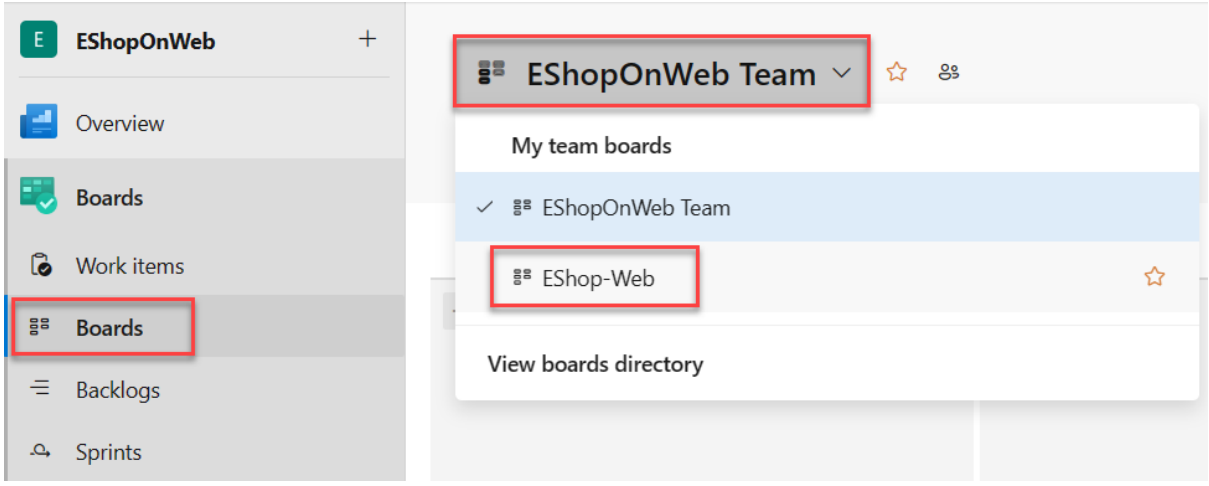
11. Select **Add link** to save the Child item

Note: On the **Training dashboard** panel, note that the assignment, **Area**, and **Iteration** are already set to the same values as the epic that the feature is based on. In addition, the feature is automatically linked to the parent item it was created from.

12. On the (New Feature) **Training dashboard** panel, select **Save and Close**.

13. In the vertical navigation pane of the Azure DevOps portal, in the list of the **Boards** items, select **Boards**.

14. On the **Boards** panel, select the **EShop-Web boards** entry. This will open the board for that particular team.

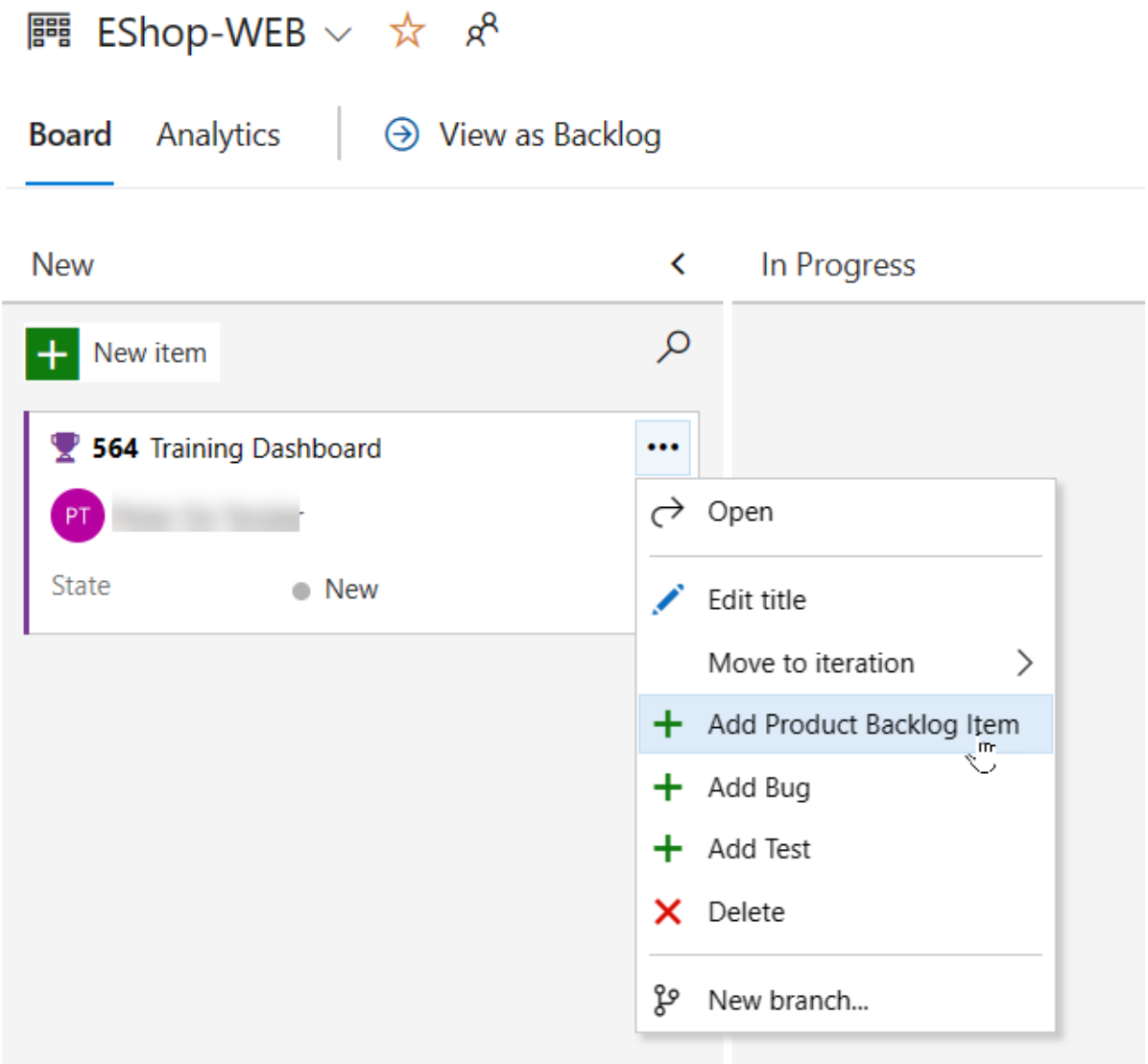


15. On the **Boards** panel, in the upper right corner, select the **Backlog items** entry and, in the dropdown list, select **Features**.

Note: This will make it easy to add tasks and other work items to the features.

16. Hover with the mouse pointer over the rectangle representing the **Training dashboard** feature. This will reveal the ellipsis symbol (…) in its upper right corner.

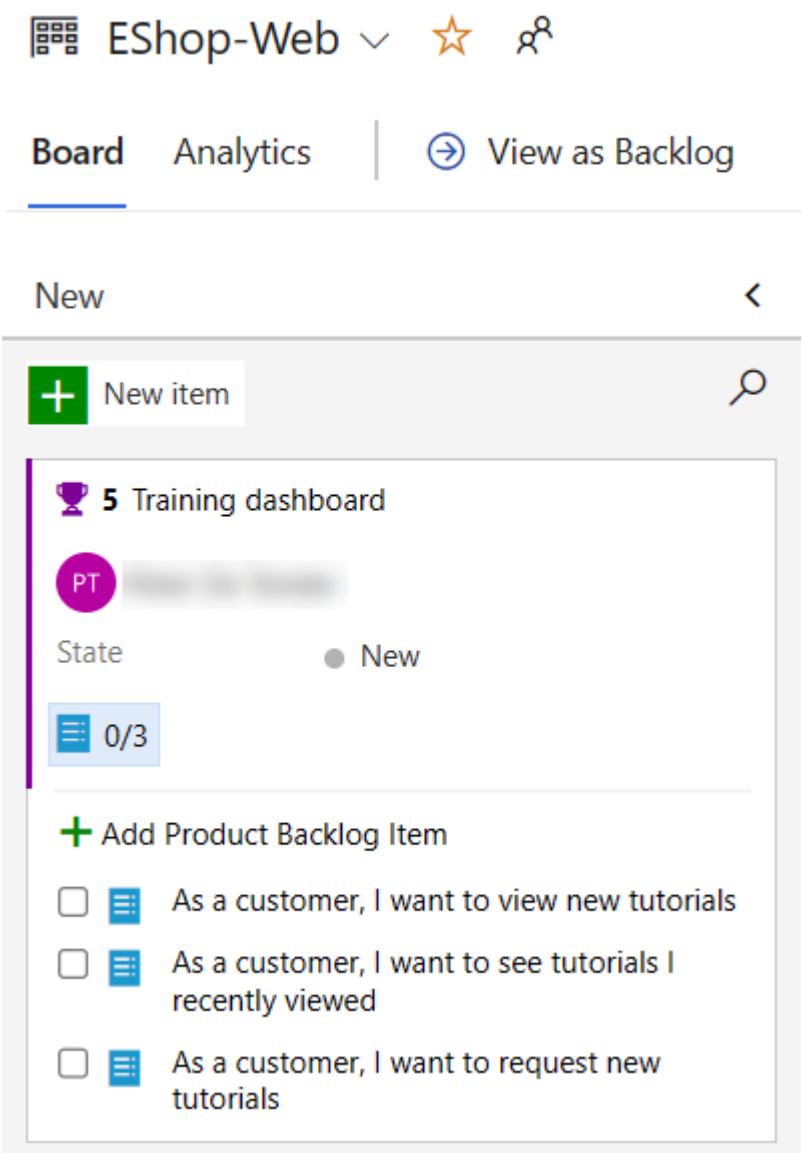
17. Select the ellipsis (…) icon and, in the dropdown list, select **Add Product Backlog Item**.



18. In the textbox of the new product backlog item, type **As a customer, I want to view new tutorials** and press the **Enter** key to save the entry.

Note: This creates a new product backlog item (PBI) work item that is a child of the feature and shares its area and iteration.

19. Repeat the previous step to add two more PBIs designed to enable the customer to see their recently viewed tutorials and to request new tutorials named, respectively, **As a customer, I want to see tutorials I recently viewed** and **As a customer, I want to request new tutorials**.



20. On the **Boards** panel, in the upper right corner, select the **Features** entry and, in the dropdown list, select **Backlog items**.

Note: Backlog items have a state that defines where they are relative to being completed. While you could open and edit the work item using the form, it's easier to just drag cards on the board.

21. On the **Board** tab of the **EShop-Web** panel, drag the first work item named **As a customer, I want to view new tutorials** from the **New** to **Approved** stage.

Note: You can also expand work item cards to get to conveniently editable details.

22. Hover with the mouse pointer over the rectangle representing the work item you moved to the **Approved** stage. This will reveal the down facing caret symbol.

23. Select the down facing caret symbol to expand the work item card, replace the **Unassigned** entry with your name, then select your account to assign the moved PBI to yourself.

24. On the **Board** tab of the **EShop-Web** panel, drag the second work item named **As a customer, I want to see tutorials I recently viewed** from the **New** to the **Committed** stage.

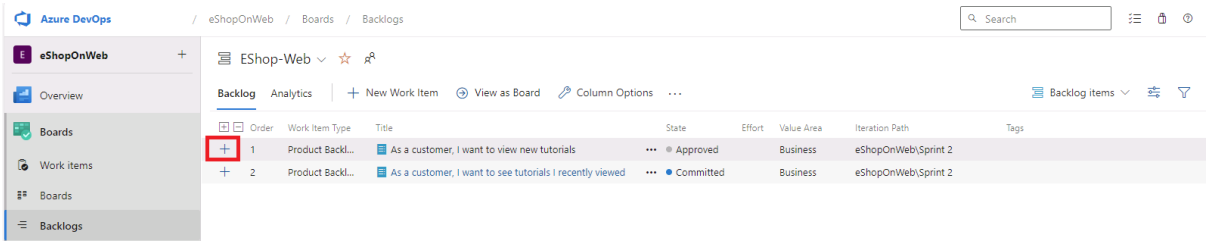
25. On the **Board** tab of the **EShop-Web** panel, drag the third work item named **As a customer, I want to request new tutorials** from the **New** to the **Done** stage.

Note: The task board is one view into the backlog. You can also use the tabular view.

26. On the **Board** tab of the **EShop-Web** pane, at the top of the pane, select **View as Backlog** to display the tabular form.

Note: You can use the second plus sign directly left to the first backlog item to add a new task to it.

27. On the **Backlog** tab of the **EShop-Web** pane, in the upper left corner of the pane, select the plus sign next to the first work item. This will display the **NEW TASK** panel.

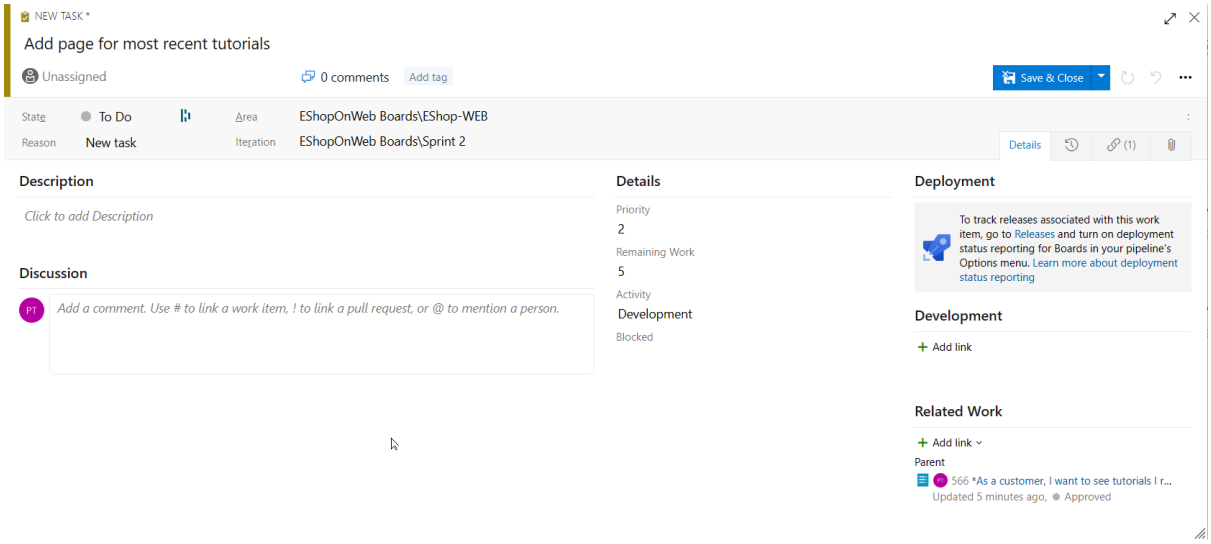


28. At the top of the **NEW TASK** panel, in the **Enter title** textbox, type **Add page for most recent tutorials**.

29. On the **NEW TASK** panel, in the **Remaining Work** textbox, type **5**.

30. On the **NEW TASK** panel, in the **Activity** dropdown list, select **Development**.

31. On the **NEW TASK** panel, select **Save and Close**.



32. Repeat the last five steps to add another task named **Optimize data query for most recent tutorials**. Set its **Remaining Work** to **3** and its **Activity** to **Design**. Select **Save and Close** once completed.